Is Sub-10-min MRI-only Comprehensive MSK Exam Clinically Feasible?

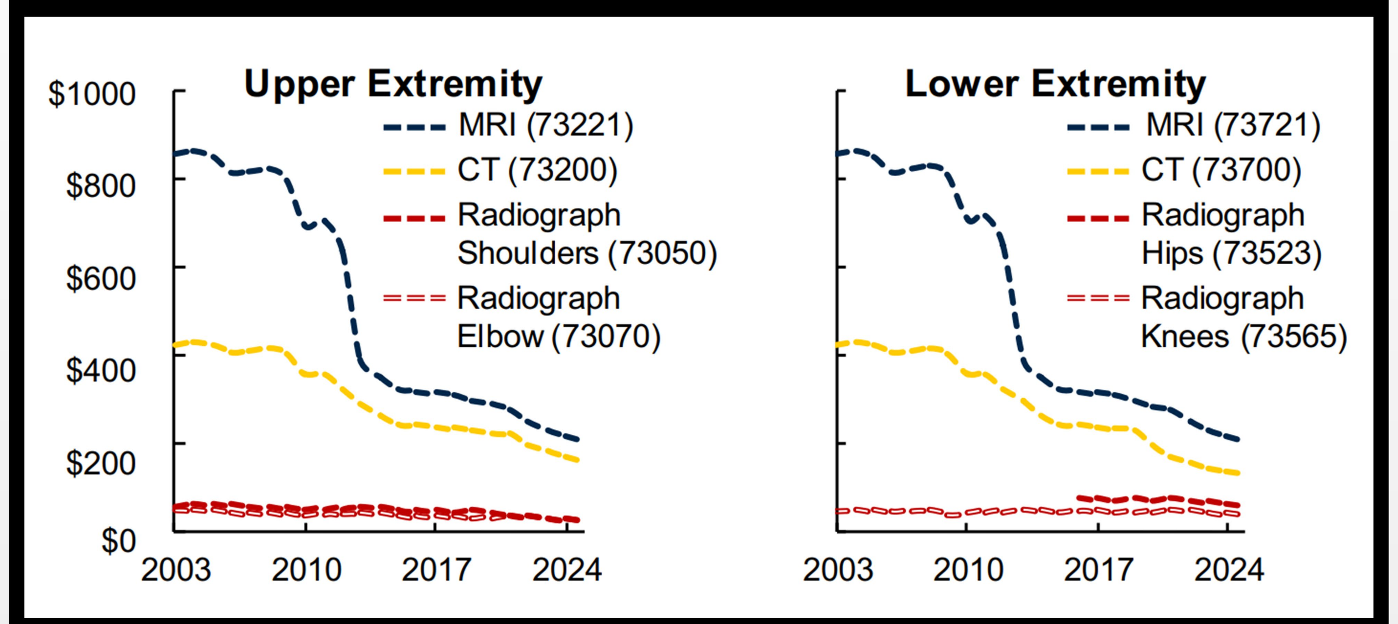
Hung Do[1], M. Bekku[1], D. Berkeley[1], M. Golden[1], S. Kitane[1], M. Uike[1], K. Shinoda[1], R. Takayanagi[1], H. Takai[1], T. Kawai[1], B. Tymkiw[1], W. AlGhuraibawi[1], K. Sorfleet[2], D. Devries[2], S. Caruthers[1], M. Kadbi[1], M. Provencher[2], S. Tashman[2], C. Ho[2]

[1]Canon Medical Systems; [2]The Steadman Clinic & Steadman Philippon Research Institute, Vail, CO, USA

1. Accessibilities & Economic Trends

MRI access is scarce & reimbursement is shrinking. An abbreviated high-valued MRI exam is *desired*.

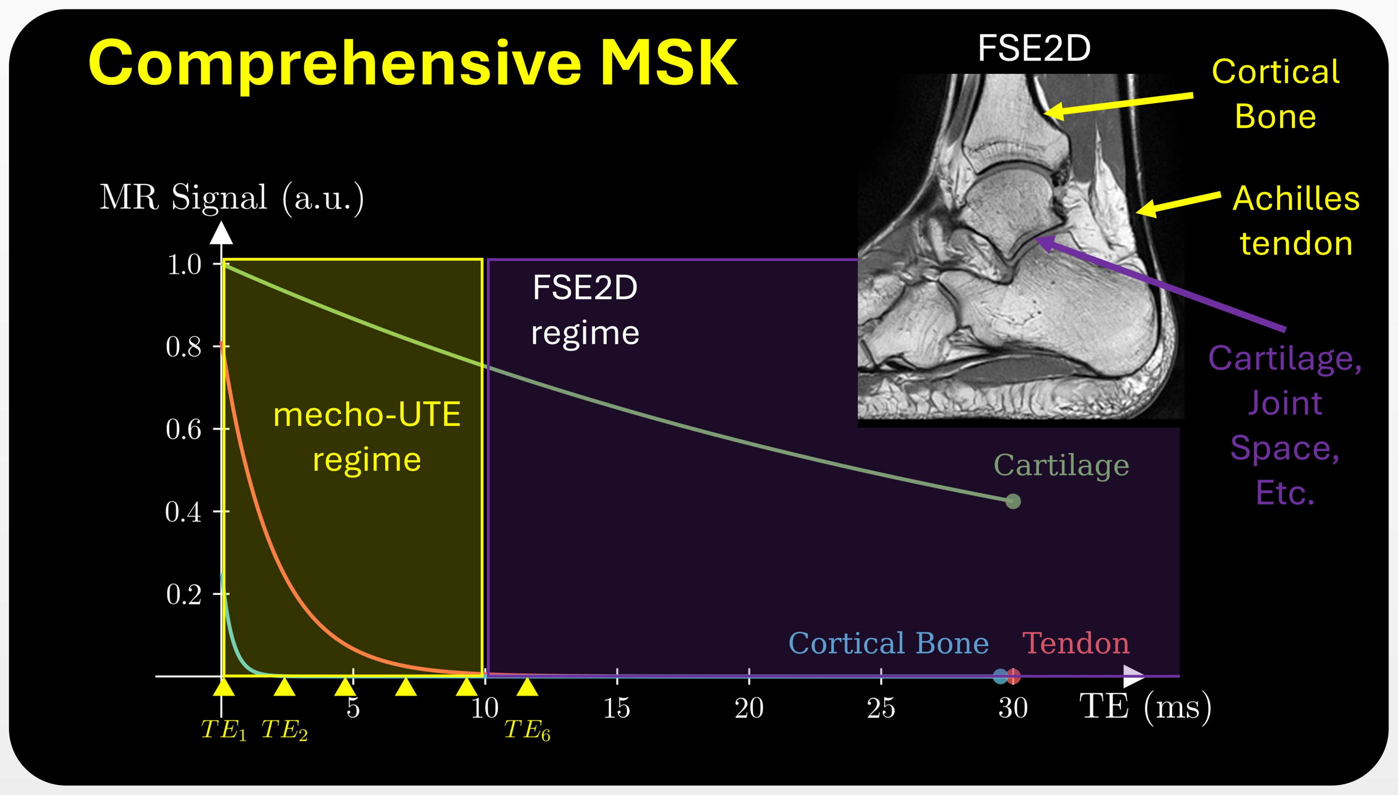
~77% Reimbursement Reduction



A. Huang et al., Clinical Imaging 2025

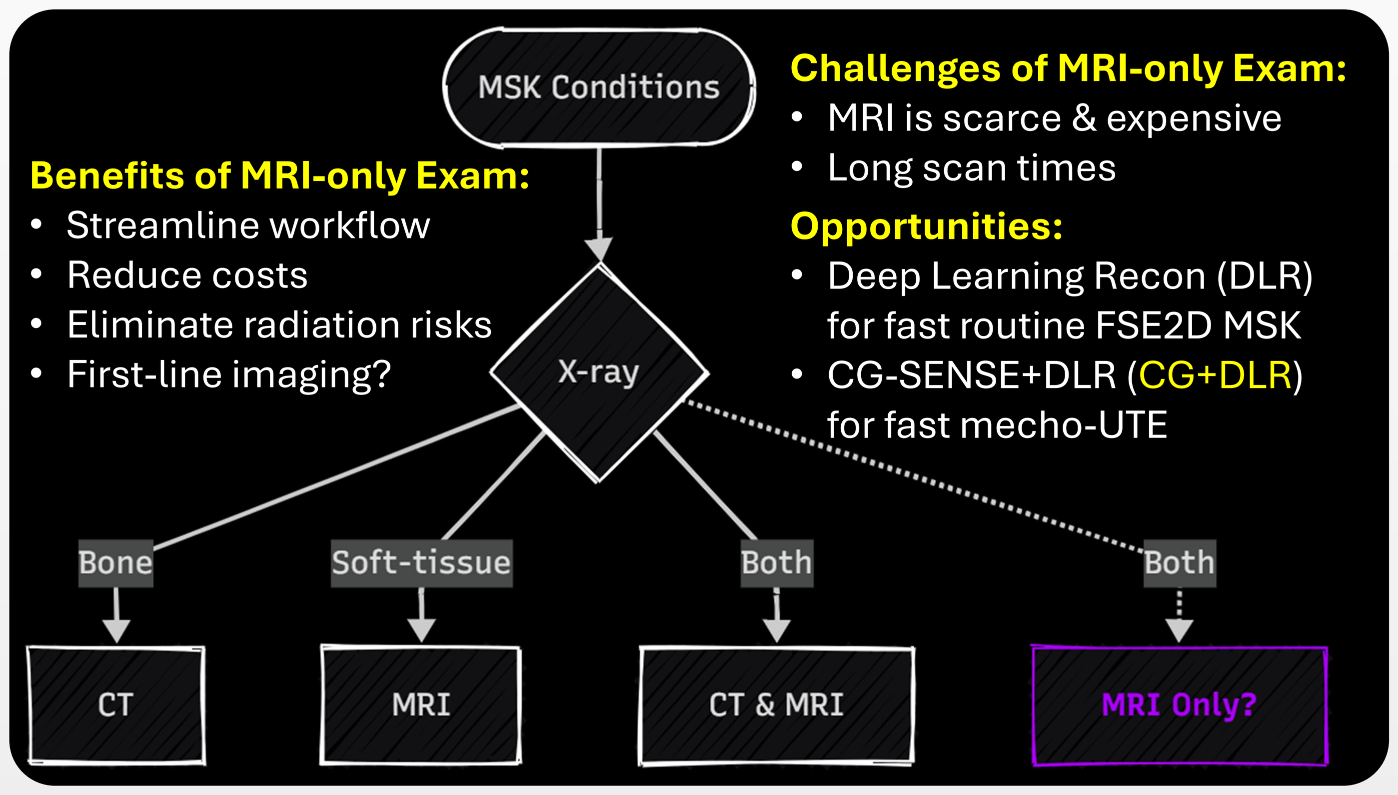
2. Gaps in Standard-of-Care MSK Imaging

Short-T2 tissues are ***invisible*** in CT and routine MRI until late disease/damage stages. Mecho-UTE helps?



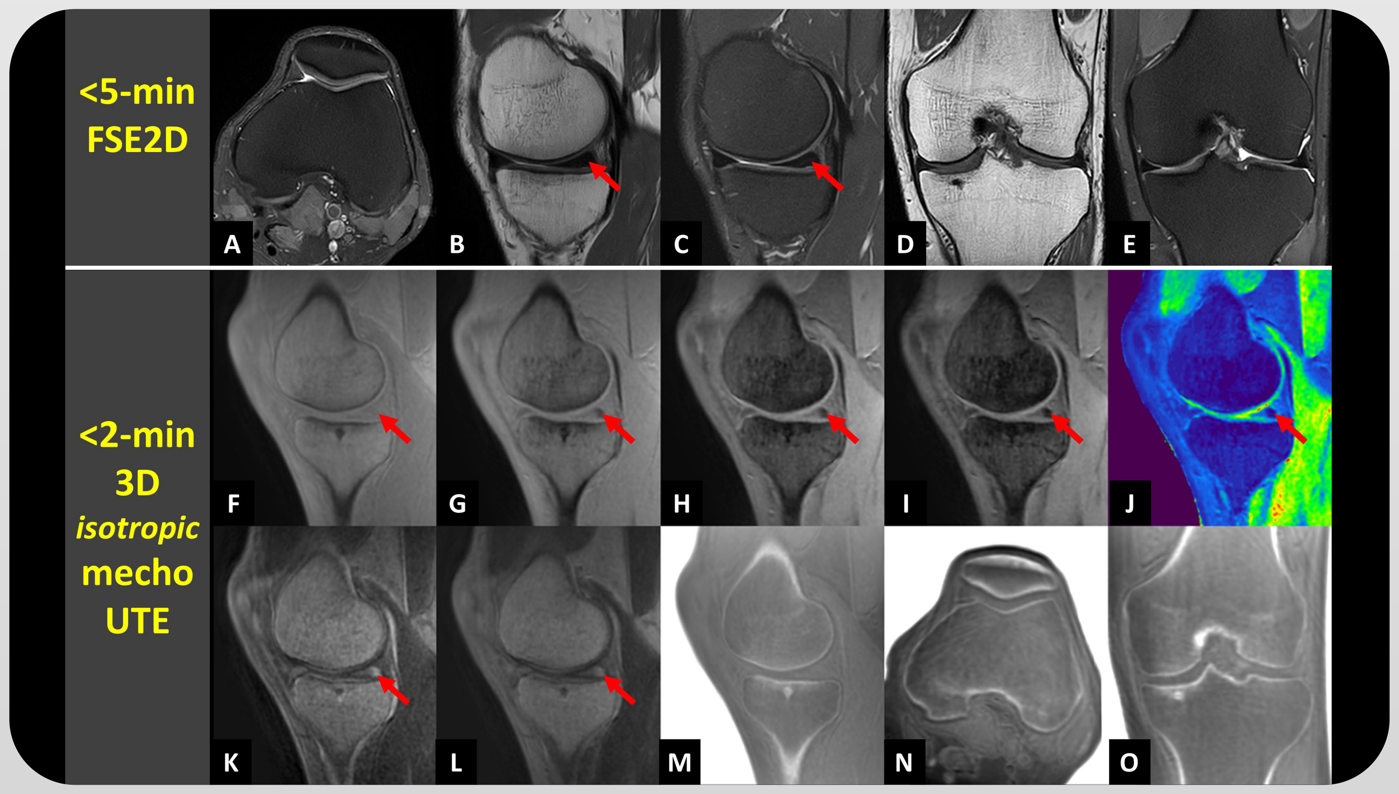
3. Open Questions & Opportunities

Can an ***MRI-only comprehensive MSK*** examination be developed and deployed clinically?



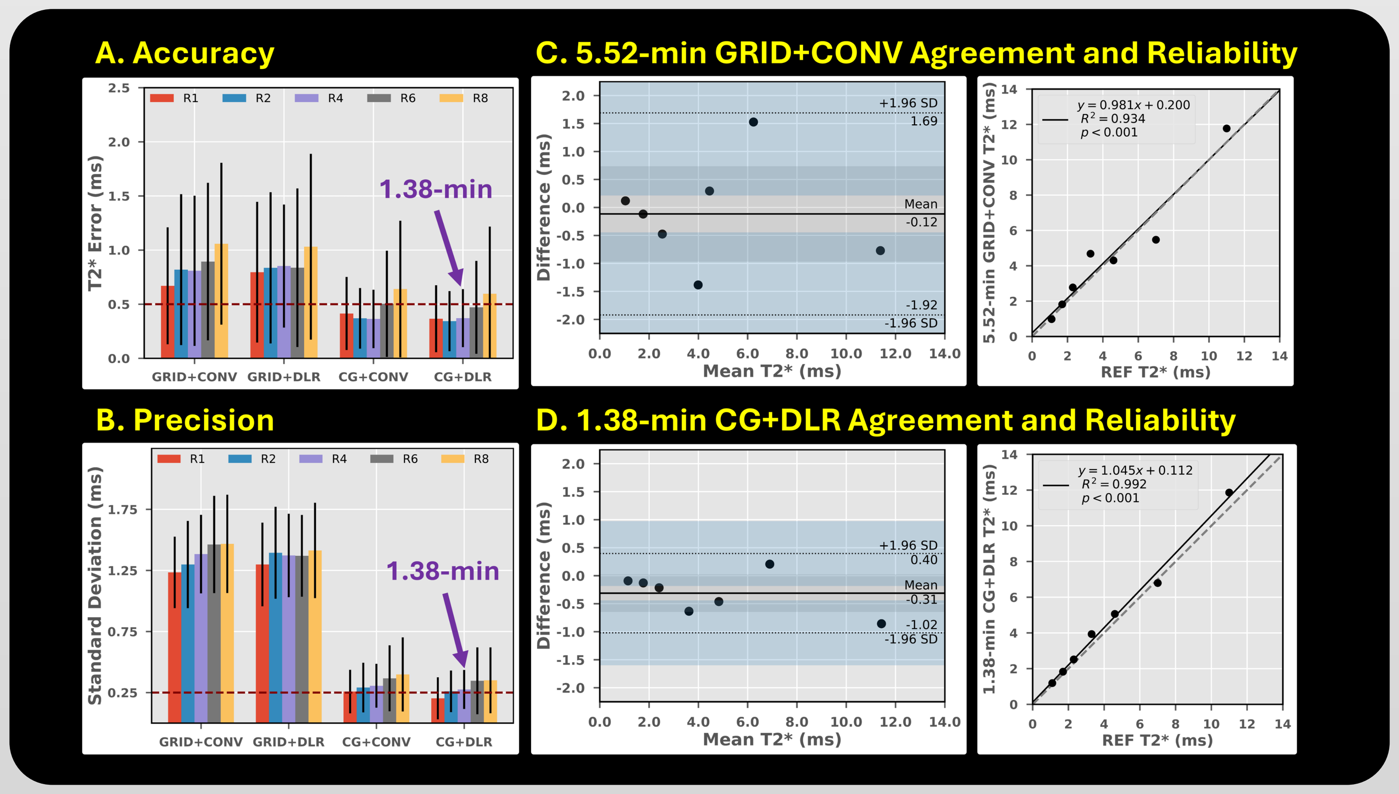
4. Proposals

DLR enables sub-8-min clinical routine MSK MRI. Sub-3-min knee MRI has been shown. We aim to achieve sub-2-min mecho-UTE with CG-SENSE+DLR.



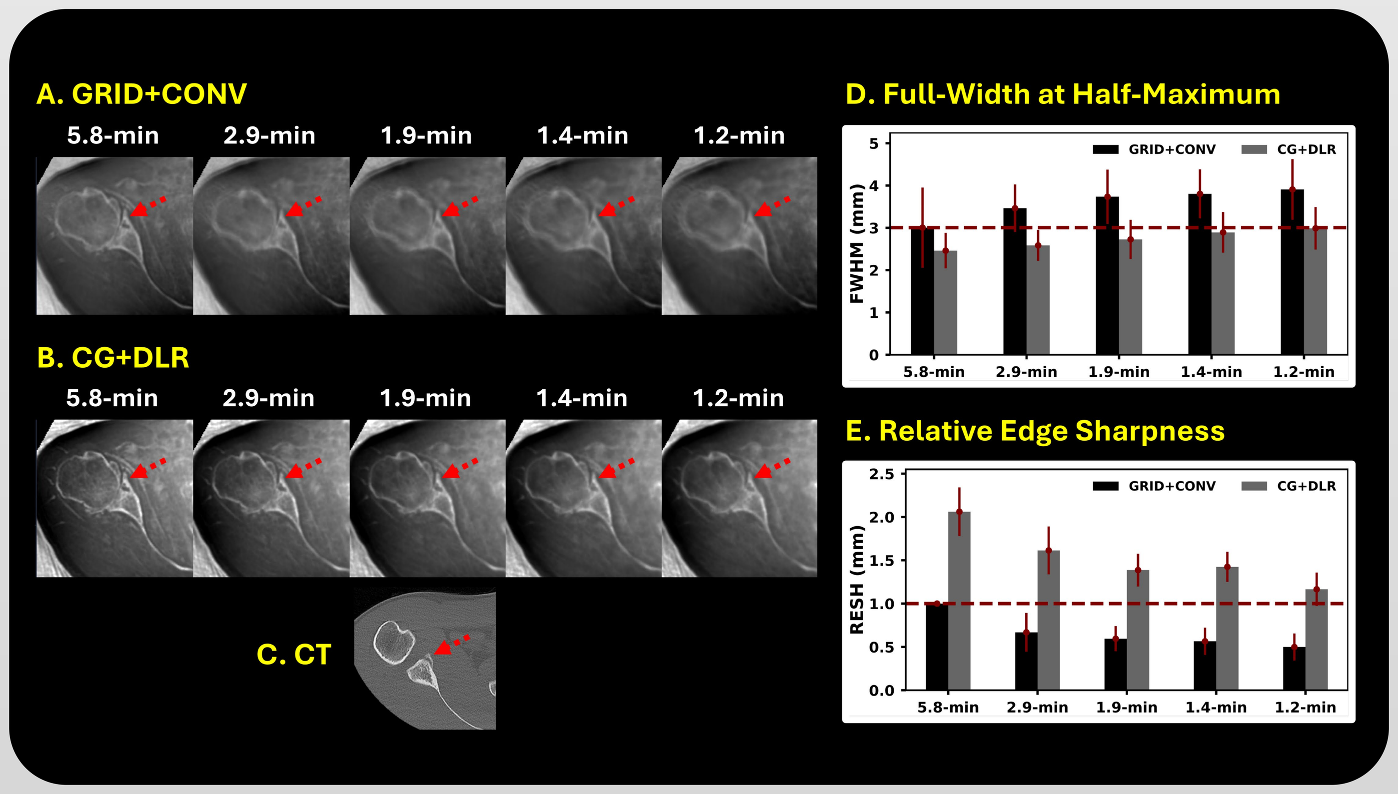
5. Phantom Evaluations

CG+DLR provides more accurate and precise T2* quantification. Sub-2-min mecho-UTE with CG+DLR has better agreement and reliability.



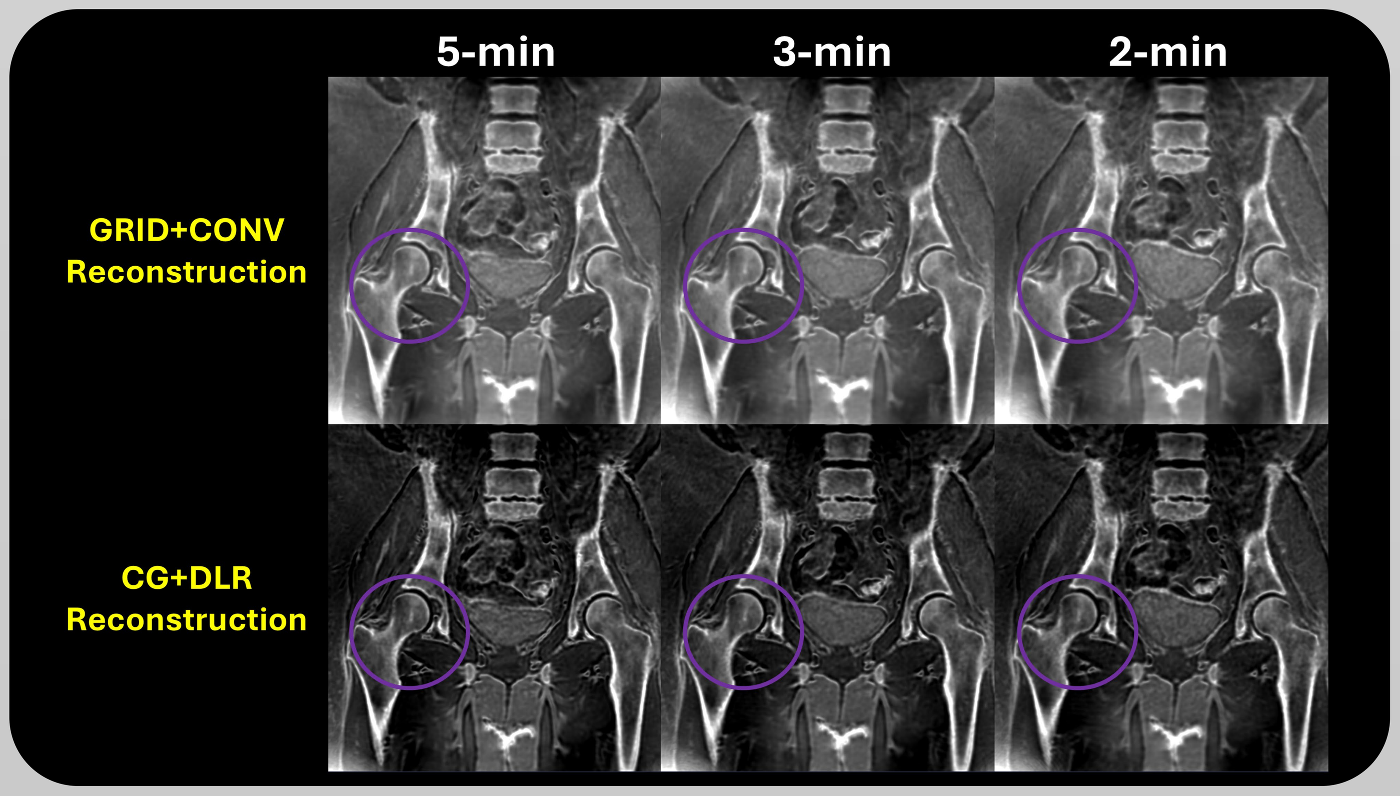
6. Retrospective *in Vivo* Evaluations

Sub-2-min mecho-UTE with CG+DLR provides better resolution and sharpness vs. 5-min mecho-UTE with GRID+CONV (i.e., gridding + conventional filter).



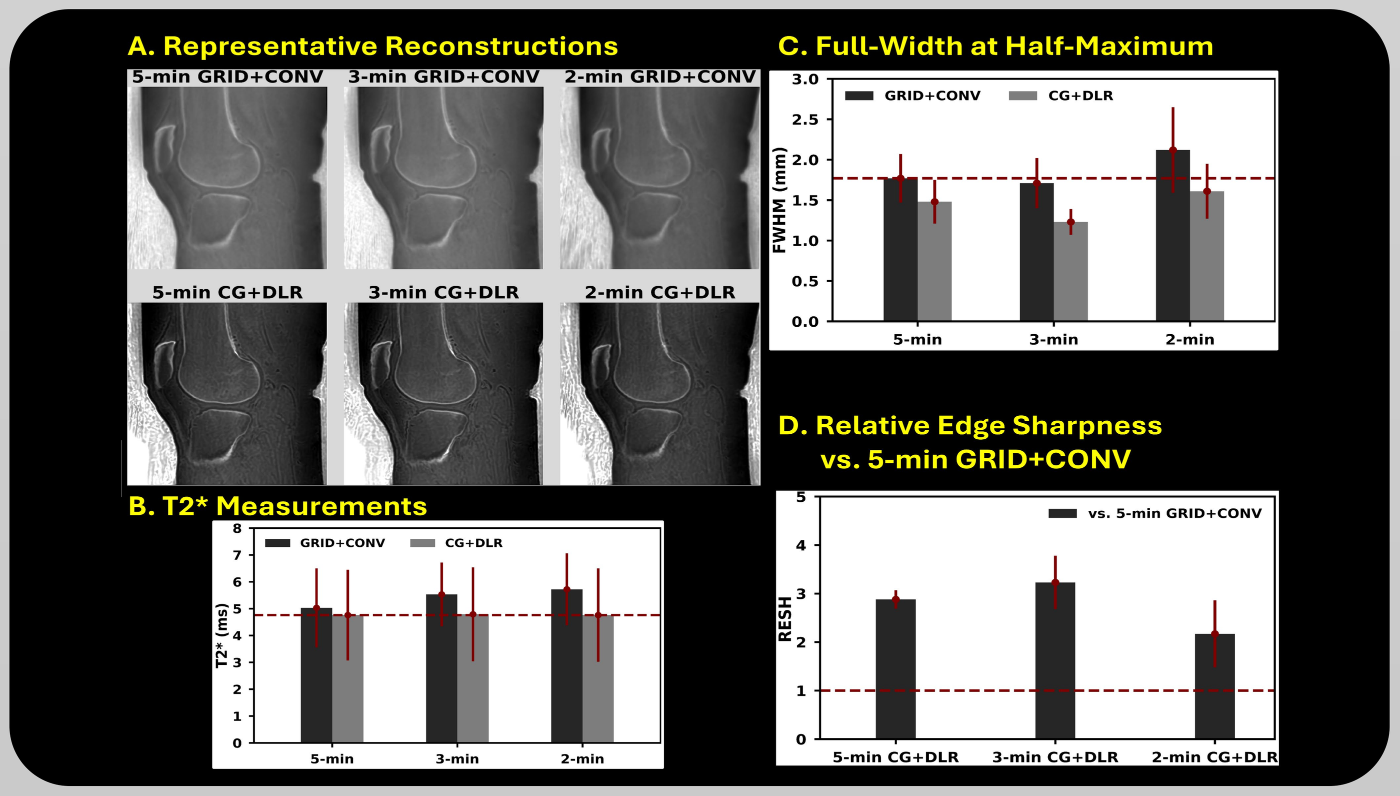
7. Prospective *in Vivo* Images

CG+DLR provides higher resolution and sharpness compared to the GRID+CONV counterparts.



8. Prospective *in Vivo* Evaluations

2-min mecho-UTE w/ CG+DLR has better resolution and sharpness vs. 5-min one with GRID+CONV.



9. Conclusions & Discussions

This preliminary study demonstrates that:

- CG-SENSE & DLR allow sub-2min mecho-UTE for simultaneous CT-like bone-weighted imaging and accurate T2* quantification of short-T2 tissues.
- Adding to an DLR-accelerated sub-8-min routine FSE2D protocol, an MRI-only comprehensive MSK exam can be performed in under 10 minutes.
- Abbreviated FSE2D MRI + 2-min mecho-UTE may be possible in under 5 minutes making it cost-effective enough for first-line imaging (*when appropriate*).
- Large clinical trials are warranted to confirm potential benefits hypothesized by this study.

Canon

Comments & Feedback

Please contact **Hung Do** at hdo@us.medical.canon

THE STEADMAN CLINIC AND STEADMAN PHILIPPON RESEARCH INSTITUTE